Day 010 — Joint and Payload Hazards

Assistive Systems for Trusted Robotic Operations

R0.3 · 2026-11-02 · Owner: Faisal Farouq Day 10 result: PASS (desk work: hazard list and control ideas). Severity, likelihood and residual risk stay UNASSESSED for a qualified reviewer.

Full table: day_010_motion_payload_hazards.csv (15 hazards, MP-01 to MP-15). On Day 13 they get permanent IDs (ASTRO-H-21 onward).

The short version

- The drives are strong enough to crush a hand. A knee at its rated torque pushes about 760 N at 10 cm from the joint, and a hip-roll joint about 1,140 N. Force limits can't make that safe, so the controls are guards, gaps and distance.
- A runaway joint is fast. At no-load speed a hip-roll joint turns 7 rad/s, so the leg tip could move about 6 m/s. That's why speed limits, a watchdog and an independent e-stop come before any motion, and why first tests use a restraint.
- The handling stand is low-energy but still pinches. The object weighs 19.6 N, but the actuator's push force can be hundreds of newtons, so the gap and the actuator choice matter more than the object.
- People will want to touch it. The pinned arms and tray look like handles. Briefings, barrier signs and the keep-out zone deal with that.

The 15 hazards

ID	Where	Hazard	Key number	Control idea	Check
MP-01	Knee	Finger or hand in the knee-pad gap	760 N rated, 2,280 N peak at 0.10 m	Guard or bellows; else gap ≥ 25 mm (fingers)	Days 97, 317
MP-02	Ankle	Finger between shin and foot cover	400 N rated, 1,200 N peak at 0.10 m	Same as knee	Days 95, 97, 317
MP-03	Hip	Hand between thighs or thigh and pelvis	1,140 N rated, 3,420 N peak at 0.10 m	Guard; else gap ≥ 100 mm (hand)	Days 93, 97, 317
MP-04	Foot	Person's foot stepped on	up to 539.55 N through one foot	Keep-out zone, barrier	Days 136, 261
MP-05	Swinging leg	Leg strikes a person	≈0.45 m/s planned foot speed	Keep-out zone; speed limits	Days 131, 136
MP-06	Any joint	Runaway joint	20.9 / 10.5 / 7.0 rad/s no-load; leg tip ≈6 m/s	Firmware + file speed limits, watchdog, independent e-stop, end stops, restraint	Days 131, 221, 233
MP-07	Shoulder pin	Pinch when posing; arm drops if pin lost	arm ≈1 kg (assumed)	Captive pin with clip; preflight check	Days 87, 261
MP-08	Shell seams	Panels collide, crack or pinch skin	—	Swept-clearance study; rounded edges	Days 97, 309, 317
MP-09	Cables	Cable snag cuts a sensor or power line	detect sensor loss ≤ 100 ms	Service loops; bend check; stale-sensor detection	Days 97, 216
MP-10	Tray	Dropped object	21.6 J to the floor from 1.1 m; 2.9 J onto a catch tray 0.15 m below	Lip + pin; catch tray; unbreakable object	Days 201, 298
MP-11	Handling actuator	Hand pinched by the tray	push force = datasheet (often 100s of N) vs 24.5 N held	Lowest push force that still holds ≥ 98 N; guard or ≥ 100 mm gap; limit switches	Days 187, 201, 299
MP-12	Stand	Stand tips over	7.35 N m; 10 kg stand needs base half-width > 0.075 m	Wide base or bolted	Day 201
MP-13	Handling power loss	Tray drops when power cut	24.5 N held; ≥ 98 N holding screen	Holding force or brake; power-off test	Days 296, 299
MP-14	Arms, tray, stand	Used as a handhold	not designed to hold a person	Briefing, sign, no contact	Days 139, 333
MP-15	Mode	Walking with the object or docked to the stand	—	Payload and support interlocks block walking	Days 237, 299

Gap values come from ISO 13854 (minimum gaps to avoid crushing parts of the body: fingers 25 mm, hand 100 mm). Treat them as starting values; the mechanical reviewer confirms them.

How the numbers were worked out

- Pinch force = joint torque ÷ distance from the joint axis. Example: 76 N m ÷ 0.10 m = 760 N. Rated values from Day 6: 40 N m direct, 76 N m at 2:1, 114 N m at 3:1. Peak values: 120, 228, 342 N m.
- Runaway speed: RobStride 04 no-load 200 rpm = 20.9 rad/s; ÷2 = 10.5; ÷3 = 7.0 rad/s. Leg tip at 0.86 m: 7.0 × 0.86 ≈ 6.0 m/s.
- Planned swing: 0.144 m in 0.6 s with a smooth profile peaks at 1.875 × the average speed: 1.875 × 0.24 = 0.45 m/s.

• Drop energy = m g h: 2 × 9.81 × 1.1 = 21.6 J; 2 × 9.81 × 0.15 = 2.9 J.

How this links to the existing register

Existing hazard	New detail added
ASTRO-H-02 pinch / shell	MP-01, 02, 03, 07, 08, 11
ASTRO-H-03 runaway	MP-06, 09
ASTRO-H-05 payload / misuse	MP-10 to 15
ASTRO-H-17 power loss	MP-13
ASTRO-H-20 people get close	MP-04, 05, 14

Outside help

A mechanical reviewer confirms the gap sizes and guard designs (Days 97 and 199) and the handling holding force (Day 296). A controls or safety reviewer confirms the runaway protections (Day 129). No one is named yet (D-02).

Checks

Check	Result
Every hazard has a control idea, a test ID and a reviewer role	PASS
No gap size, force limit or residual risk presented as approved	PASS
Person lifting and walking with cargo listed as excluded uses	PASS (MP-14, MP-15)

Sources: R0.1 hazard register ASTRO-H-02, 03, 05, 17, 20; R0.1 verification register; Day 6 R0.3 drive data; Day 9 keep-out sketch; ISO 13854 (minimum gaps to avoid crushing; values to be confirmed by the reviewer).